

What Exactly is a Linux Distribution?

If you ever read the IT press, you may have heard of the larger Linux company names such as Red Hat (RHEL/CentOS/Fedora), Canonical (Ubuntu) and SuSE. There are literally thousands of other smaller companies and organisations that also make Linux distributions. Examples of which can be seen on websites like [distrowatch](http://distrowatch.org).

These are all companies or organisations that have created their own ‘distributions’ or flavours of Linux, there are distributions to cater to everyone’s needs. Probably too many distributions! In the Ultimate Linux Newbie Guide we will keep it simple by basing it only on the most popular distributions.

In any distribution, the fundamentals stay the same:

- There is always a Linux Kernel. This is the core component of the Linux operating system which Linus Torvalds wrote back in 1991 (It’s got thousands of contributors now!). The kernel is the interface between the hardware on your computer (keyboards, mice, displays, etc) and the software.
- The default GNU software (tools like bash – ls, rm, etc). These are mainly command-line based utilities which make a core (but critical) part of any UNIX system. Think of it as if the kernel is the bus between the hardware and the software, think of these like the tool-kit you need to keep the bus on the road!
- General software to be expected of a desktop Linux distribution. Usually this would include software such as text editors, web browser, email client and probably a word processor or office suite, etc.

What differs from distribution to distribution?

- General software: (Office Apps, Prog. Languages, Games, Graphics software, etc). Some distributions are built to be lightweight and deliberately don’t ship with much software. Some are quite the opposite. It all depends upon what you need as a user. Most of the time you can install more software very easily if it isn’t pre-bundled (more on this in a later chapter)

“Some distributions are built with older computers in mind, such as [Lubuntu](#) (left) and [Bodhi Linux](#) (right). The quantity and quality of the software pre-bundled can be varied. This is often because corners have been cut to ensure the desktop experience runs smoothly on PCs that have as little as 512MB RAM and a 700MHz CPU. That’s less than the minimum requirements for Windows 7!”

- **Cost** – whether you pay nothing, a little, or a lot for a distribution depends on what you need from it and the business model the distributor works to. For example, Red Hat Enterprise Linux (RHEL) and SuSE are both commercial Linux distributions which have business level support contracts. They offer telephone hotlines and round-the-clock service level based support for businesses that rely on their Linux systems right up to mission-critical needs. Note however, that just because you have paid for the software, doesn’t necessarily mean that it’s better. You are paying for licensing, documentation, salaried staff, end-user feature enhancements and telephone support and other matters of that nature
- **Quality of software** – (buggy or not buggy software, latest versions of software)

- **Documentation and Manuals** – (Quality of, Lack of, Quantity of)
- **Installation Software** – Distributions often use different methods of ‘packaging up’ their software. For Red Hat based systems like RHEL, CentOS, Fedora and SuSE, this is the RPM standard (used by tools such as YUM and DNF). Debian based systems like Ubuntu and Linux Mint use one called APT. Effectively, this is mostly irrelevant these days, however they still work in slightly nuanced ways
- **Current** – Some distributions go months or even years without being updated. They might consider the release ‘stable’ and therefore don’t provide an update unless it’s a major security fix, or until they are good and ready to update. Some distributions, generally the more niche ones also wind up and shut down.
- **Support** – Many of the more common distro’s have well established community support on web forums or chat channels). Some of them also offer a free Linux installation and software, but you can also opt-in for a support contract (Ubuntu offers this)
- **Ease of Use** – How easy it is to use overall.

So in summary, whatever distribution you choose you may get a better range of options with distribution X over distribution Y. However Y may suit your specific needs more than X in some cases. The choice is for you to decide. As Linux is almost always free to download, it lends itself to be evaluated until you find just the one you are looking for.